

Association Between American Board of Surgery Initial Certification and Medical Malpractice Payments

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Objective: To measure associations between surgeons' examination performance and obtaining American Board of Surgery certification with the likelihood of having medical malpractice payments.

Background: Further research is needed to establish a broader understanding of the association of board certification and patient and practice outcomes.

Methods: Retrospective analysis using propensity score-matched surgeons who attempted to obtain American Board of Surgery certification. Surgeons who completed residency between 2000 and 2019 (n = 910) and attempted to become certified were categorized as certified or failing to obtain certification. In addition, groups were categorized as either passing or failing their first attempt on the qualifying and certifying examinations. Malpractice payment reports were dichotomized for surgeons who either had a payment report or not.

Results: The hazard rate (HR) of malpractice payment reports was significantly greater for surgeons who attempted and failed to obtain certification [HR = 1.87; 95% confidence interval (CI), 1.28–2.74] than for surgeons who were certified. Moreover, surgeons who failed either the qualifying (HR = 1.64; 95% CI, 1.14–2.37) or certifying examination (HR = 1.72; 95% CI, 1.14–2.60) had significantly higher malpractice payment HRs than those who passed the examinations on their first attempt.

Conclusions: Failing to obtain board certification was associated with a higher rate of medical malpractice payments. In addition, failing examinations in the certification examination process on the first attempt was also associated with higher rates of medical malpractice payments. This study provides further evidence that board certification is linked to potential indicators for patient outcomes and practice quality.

Keywords: board certification, medical malpractice payments, assessment (*Ann Surg* 2022;276:281–287)

The American Board of Surgery (ABS) is a standard-setting organization that issues certificates to surgeons who have met defined standards of training, education, knowledge, and

judgment.¹ After meeting all prerequisites, the final stage in the initial certification process requires passing a multiple-choice qualifying examination (QE) as well as an oral certifying examination (CE). Board certification is intended to be one indicator of a surgeon's commitment to providing quality care. While board certification is voluntary (ie, the state does not mandate board certification for practice), many hospitals require board certification for credentialing. Hospitals, insurance providers, and patients all rely on board certification as an indicator of quality care.^{2–4}

As the public relies on the value of board certification, it is important to provide evidence that board certification relates to practice outcomes. Previous research has shown that surgeons who obtain initial certification and maintain certification in surgery are less likely to have severe disciplinary actions than those who do not.^{5–7} Similar studies linking board certification and examination performance to disciplinary actions can be found in a variety of disciplines.^{8–12} While disciplinary actions are an important practice outcome, they are a somewhat broad criterion to measure, as they encompass issues that may or may not relate to patient care (eg, engaging in fraudulent or criminal activity). Critics of board certification argue that research needs to specifically address the link between board certification and patient care.¹³

Research in other specialties has linked initial board certification status and examination performance to patient outcomes.^{14–17} In general surgery, a link between board certification and patient outcomes has also been established. First, ABS certification has been associated with reduced patient morbidity and mortality in segmental colon resection.¹⁸ In addition, early-career general surgeons who were board-certified had lower morbidity and mortality rates in performing partial colectomies than early-career general surgeons who did not obtain board certification.¹⁹ Each of these studies, while important, have focused only on a single type of outcome.

We sought to build on previous research by analyzing whether board certification would be associated with adverse patient care outcomes beyond those limited to specific procedures. Specifically, the purpose of this study was to evaluate the relationship between board certification status and examination performance with malpractice payments. While some may argue that medical malpractice payments are unrelated to the quality of care, recent research has shown a link between patient outcomes and medical malpractice payments. Specifically, variation in adverse events from patient safety indicator data have been associated with medical malpractice payments at a hospital level.^{20,21} In addition, the number of medical malpractice payments for a physician has been linked to the number of patient complaints a physician receives, suggesting that claims are not arbitrary.²² Other research has shown that the overwhelming majority of paid malpractice claims are associated with evidence of medical errors and that payments that lacked evidence of

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errors were much less common.²³ However, a systematic review of the research found mixed results for the relationship between malpractice liability and health care quality, with some studies not establishing a link between quality and malpractice payments.²⁴ Thus, the relationship between malpractice cases and quality remains open to question. Understanding this, we nevertheless hypothesized that surgeons who failed to obtain board certification in general surgery would have higher rates of malpractice payments. In addition, we also hypothesized that surgeons who had to repeat examinations (either the QE or CE) would have higher rates of malpractice payments.

METHODS

Surgeon Sample

Certification, examination performance, and demographic data were collected from the ABS database for all surgeons who completed a residency in the database through 2019. Surgeon data included sex, international medical school graduation status, and examination performance (whether the surgeon had to repeat the QE or CE). Surgeons must first pass the QE before they can attempt the CE. The ABS data were then joined with publicly available Centers for Medicare & Medicaid Services (CMS) data from 2012 to 2016 to account for surgeon practice variables. The public CMS data includes data for the number of beneficiaries seen each year (ie, practice volume) as well as the average beneficiary risk score (the expected average health care costs for their patients) for a surgeon provider. Finally, we also joined median household income from the US Census Bureau²⁵ to the location of the surgeon's practice by zip code due to potential differences in the likelihood of board-certified and noncertified surgeons having patients of different socioeconomic status, as a link between the likelihood of filing a malpractice claim has previously been linked to patient income levels.²⁶

Before joining the combined ABS/CMS data with the malpractice data, we balanced the characteristics of the control (certified) and the treatment (noncertified) groups by matching propensity scores a priori, which were calculated using R statistical software, version 4.0.3, and the MatchIt package (version 3.0.2) package (R Foundation for Statistical Computing). This method utilized logistic regression to estimate the likelihood of being in the treatment group, based on surgeon covariates with the nearest neighbor matching without replacement. Neither the average patient beneficiary risk score nor median household income were significantly different between the noncertified and certified groups and were therefore excluded from the propensity score-matching process. The final propensity score-matched data included a 5:1 ratio of certified to noncertified surgeons on sex, international medical school graduation status, and number of beneficiaries.

The full ABS records, which included all surgeons as well as an indicator for the matched surgeons from the propensity scoring process were shared with the Health Resources and Services Administration (HRSA) who merged malpractice data from the National Practitioner Data Bank (NPDB) onto the ABS data. The NPDB is a web-based repository of reports containing information on medical malpractice payments and certain adverse actions related to health care practitioners, providers, and suppliers. Established by Congress in 1986, it is a workforce tool that prevents practitioners from moving across states without disclosure or discovery of damaging performance.²⁷ Records could only be reliably linked for graduates of residency programs from 2000 and later, resulting in the removal of any surgeons who

completed residency before 2000. The NPDB data only includes records for surgeons who had a malpractice payment. The combined data were deidentified and sent back to the ABS for analysis. By using propensity score matching, we were able to match surgeons on covariates and still have the ABS analyze a deidentified dataset that did not include variables which could possibly reidentify surgeons (which would not have been possible if we had used a regression model with covariates). The ABS staff who analyzed the data had no ability to individually identify a surgeon's malpractice data with their ABS data. This research was determined to be exempt from review by the Advarra Institutional Review Board.

Outcomes Measures

The NPDB data included number of malpractice payment reports, patient outcome, and the year of the malpractice payment. Patient outcomes included death, emotional injury only, insignificant injury, major permanent injury, major temporary injury, minor permanent injury, minor temporary injury, quadriplegic, brain damage, lifelong care, significant permanent injury, and finally injuries that could not be classified based on the available data. These patient outcomes were combined into a dichotomous outcome of either having a malpractice payment report or not. Surgeons with any medical malpractice payments had an outcome coded as 1, whereas surgeons without a malpractice payment record had an outcome coded as 0.

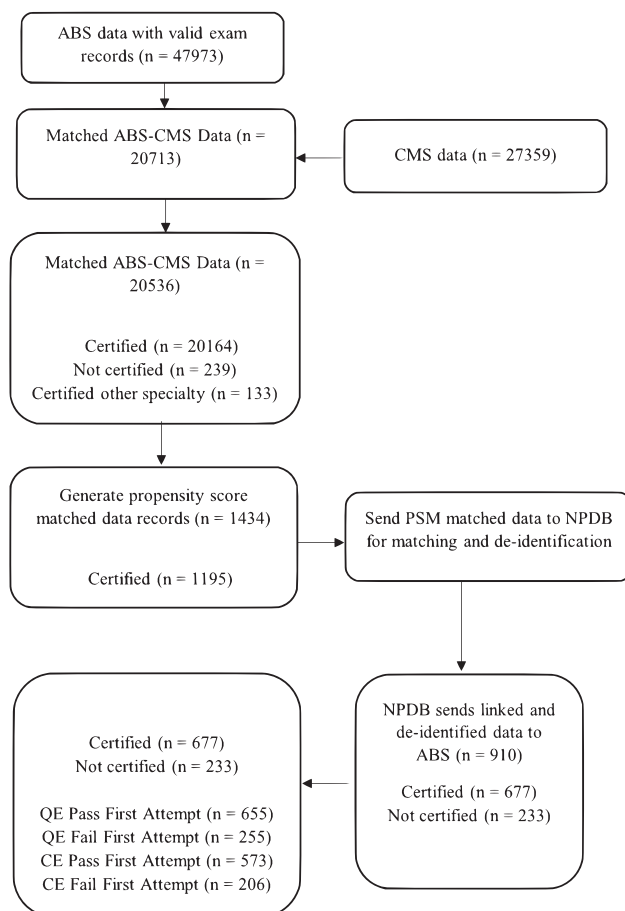


FIGURE 1. Flow diagram for creation of final analysis data. PSM indicates propensity score matching.

TABLE 1. Demographic Characteristics and Malpractice Injury Incidence Rates for Certification and Examination Performance Groups for the Merged ABS/NPDB Data

Characteristics	Obtained Certification (n = 18,505)	Did Not Obtain Certification (n = 1875)	Passed First QE (n = 17,661)	Failed First QE (n = 2719)	Passed First CE (n = 15,801)*	Failed First CE (n = 3412)*
Full sample (N = 20,380)						
Sex						
Male	13,021	1309	12,468	1862	11,026	2484
Female	5484	566	5193	857	4775	928
Medical school†						
USMG	15,455	1477	14,752	2180	13,293	2712
IMG	3040	397	2899	538	2498	700
Person-years at risk	171,581	12,873	157,541	26,913	145,934	29,837
Malpractice payment	2311	150	1975	486	1822	537
Incidence per 1000 person-years (95% CI)	13.5 (12.9–14.0)	11.7 (9.8–13.5)	12.5 (12.0–13.1)	18.1 (16.5–19.7)	12.5 (11.9–13.1)	18.0 (16.5–19.5)
Propensity score–matched sample (n = 910)						
Sex						
Male	394	157	380	171	328	137
Female	283	76	275	84	245	69
Medical school						
USMG	523	182	513	192	445	158
IMG	154	51	142	63	128	48
Person-years at risk	6419	1788	6011	2196	5422	1738
Malpractice payment	85	39	78	45	65	34
Incidence per 1000 person-years (95% CI)	13.1 (10.3–15.8)	21.8 (15.0–28.7)	13.0 (10.1–15.9)	20.5 (14.5–26.5)	12.0 (9.1–14.9)	19.6 (13.0–26.1)

*Fewer surgeons took the CE due to not passing the QE or voluntary withdrawal.

†Medical school records were missing for 11 candidates.

IMG indicates international medical graduate; USMG, United States medical graduate.

Primary Analyses

Malpractice report rates were compared for surgeons who obtained board certification with surgeons from both the full and the propensity score–matched sample who did not obtain board certification and who had taken the QE at least 1 time. Moreover, the relationship between examination performance and malpractice report incidence rates was evaluated by comparing malpractice rates for surgeons who passed both the QE and CE on their first attempt to surgeons who had repeat attempts on the QE or the CE, regardless of certification status. Malpractice rates were calculated using 1000 person-years at risk to control for exposure. Any inferences made from the full sample were limited due to a lack of control for covariates. Therefore, a Cox proportional hazards regression was used in the propensity score–matched sample (so that surgeon covariates could be properly accounted for) to compare malpractice rates for certified and noncertified surgeons as well as those who failed their first attempt on an examination to those who passed both examinations on their first attempt. Failing the QE on the first attempt and failing the CE on the first attempt were analyzed in different regression models, as passing the QE is a prerequisite to passing the CE. If the data were combined into one regression model, candidates who never took the CE would have been removed from the QE analysis. By analyzing each examination group separately, the independent utility of the examinations were evaluated. Moreover, the relationship between QE and CE scores is fairly weak, further reinforcing the utility of studying each examination in a separate model.²⁸ If a surgeon had

multiple malpractice reports, the most severe patient outcome was retained for analysis to illustrate the most severe cases in the sample. We used R statistical software, version 4.0.3, including the survival (version 3.2-3) and survminer (version 0.4.8) packages (R Foundation for Statistical Computing). All significance tests were set to a 2-sided α level of 0.05. Covariates were not included in the regression model because the propensity score–matching process had already matched the certified group to be similar in covariates to the noncertified group.

Sensitivity Analyses

To address potential concerns about errors in matching or having a biased sample, we assumed a 20% error rate in the match between the ABS sample and the NPDB sample for malpractice payment reports by assuming that all errors were certified surgeons who failed to be identified in the NPDB data. We, therefore, increased the percentage of payments in the propensity score–matched sample by 20% in certified surgeons who did not previously have a report. At an individual level, these reports were assigned randomly so that exposure would not be biased in the certified surgeons who were assigned malpractice payment reports. We also conducted an additional sensitivity analysis that removed minor temporary injuries, minor permanent injuries, and insignificant injuries from the data (leaving only severe injuries), to test whether any observed differences between the certified and noncertified surgeons could be explained by minor claims.

RESULTS

A total of 910 surgeons were successfully joined from the NPDB onto the ABS data after propensity score matching. Figure 1 shows the diagram and sample sizes of joining CMS data onto the ABS data, the propensity score–matching process, and finally joining the NPDB malpractice data onto the ABS data along with corresponding sample sizes for each step in the process. Table 1 shows the demographic characteristics and incidence rates for certified and noncertified surgeons as well as the examination performance groups of the full and propensity score–matched samples.

Full Sample

In the full sample (the 20,380 surgeons with at least 1 QE attempt without controls for covariates), noncertified surgeons had an incidence rate per 1000 person-years of 11.7 [95% confidence interval (CI), 9.8–13.5]. Meanwhile, certified surgeons had an incidence rate per 1000 person-years of 13.5 (95% CI, 12.9–14.0). Surgeons who failed the QE on their first attempt (incidence rate per 1000 person-years, 18.1; 95% CI, 16.5–19.7) had a higher incidence of malpractice payments than surgeons who passed the QE on their first attempt (incidence rate per 1000 person-years, 12.5; 95% CI, 12.0–13.1). Surgeons who failed the CE on their first attempt had a higher incidence (incidence rate per 1000 person-years, 18.0; 95% CI, 16.5–19.5) than surgeons who passed the CE on their first attempt (incidence rate per 1000 person-years, 12.5; 95% CI, 11.9–13.1).

Propensity Score–matched Sample

In the propensity score–matched sample, noncertified surgeons had an incidence rate per 1000 person-years of 21.8 (95% CI, 15.0–28.7). Meanwhile, certified surgeons had an incidence rate per 1000 person-years of 13.1 (95% CI, 10.3–15.9). Surgeons who failed the QE on their first attempt had an incidence rate per 1000 person-years of 20.5 (95% CI, 14.5–26.5) and surgeons who passed the QE on their first attempt had an incidence rate per 1000 person-years of 13.0 (95% CI, 10.1–15.9). Surgeons who failed the CE on their first attempt had an incidence rate per 1000 person-years of 19.6 (95% CI, 13.0–26.1) and surgeons who passed the CE on their first attempt had an incidence rate per 1000 person-years of 12.0 (95% CI, 9.1–14.9).

Figure 2 shows the frequency of the categories of malpractice payments in the propensity score–matched sample. A total of 13.2% of general surgeons had a malpractice payment. Death was the most commonly cited reason for a malpractice payment, followed by major temporary injuries, and significant permanent injuries. The categories with the fewest records were either “cannot be determined” or “insignificant injury.”

The Cox proportional hazards regression in the propensity score–matched sample showed that the hazard rate (HR) for malpractice payments for noncertified surgeons was significantly higher (HR, 1.87; 95% CI, 1.28–2.74) than for surgeons who obtained certification. The cumulative incidence plot comparing noncertified to certified surgeons is shown in Figure 3. The HR for malpractice payments for surgeons who failed their first QE attempt was significantly higher (HR, 1.64; 95% CI, 1.14–2.37) than for surgeons who passed their first QE attempt. In addition, the HR for malpractice payments for surgeons who failed their first CE attempt was significantly higher (HR, 1.72; 95% CI, 1.14–2.60) than for surgeons who passed their first CE attempt. Figures 4 and 5 show the cumulative incidence plot comparing first-time examination performance for both the QE and CE, respectively. The

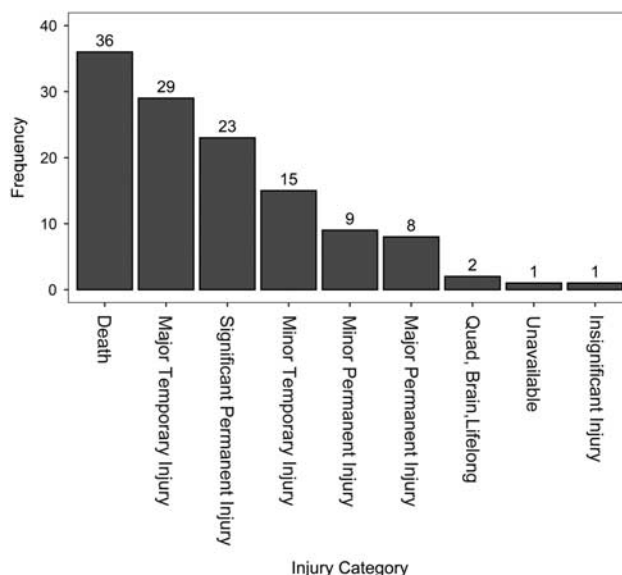


FIGURE 2. Frequencies of each category for medical malpractice payments.

assumptions that certification status and examination performance had a linear relationship with malpractice injury rates over time was tested and was not time-dependent, supporting the use of the models.

Sensitivity Analyses

The certified surgeon group in the propensity score–matched sample had 85 malpractice payments, meaning that the sensitivity analysis increased those payments by 17 (20%) to 102 total payments (with random assignment for exposure). After increasing the malpractice payment rates in the certified group, the HR for malpractice payments for noncertified surgeons was still significantly higher than certified surgeons (HR, 1.58; 95% CI, 1.09–2.30; $P=0.01$). In addition, we analyzed the sample with 25 insignificant or minor injuries removed from the data ($n=885$). The HR for malpractice payments for noncertified surgeons was significantly higher than certified surgeons when only considering more serious injury claims (HR, 1.72; 95% CI, 1.12–2.66; $P=0.01$).

DISCUSSION

This study demonstrated an association between failing to obtain board certification in general surgery and malpractice payment rates in a propensity score–matched sample of surgeons based on demographic and practice characteristics. Moreover, this study showed a relationship between examination performance (either failing the QE or CE) and malpractice injury payment rates. This study builds on the growing body of evidence that has established a relationship between obtaining board certification and practice outcomes.^{5–19} Previous research has established a link between board certification and severe disciplinary actions. However, disciplinary actions can encompass both adverse patient care events as well as behaviors that do not relate to patient care (eg, unethical behavior such as billing fraud). Other research has established the link between being board-certified and a specific procedure in general surgery (ie, partial colectomy). The unique contribution of this study is

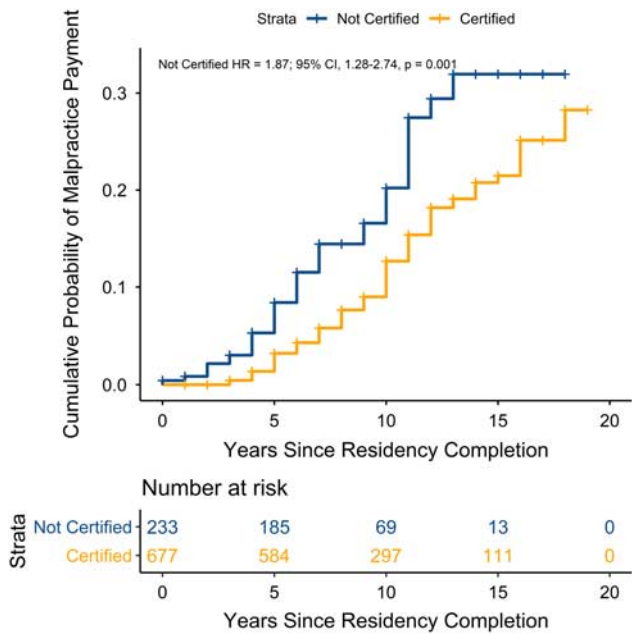


FIGURE 3. Cumulative incidence plot of medical malpractice payments over time by certification status.

that the outcome variable may be a proxy for patient care and is not limited to any specific procedure. The nature of these outcomes is not trivial, ranging from emotional injury to major permanent injuries and death.

This study provides further evidence that each step in the certification process has utility. Performance on both the QE and the CE were linked to subsequent injury payments, with those who failed either of the examinations on their first attempt having higher malpractice payment rates. The findings from this

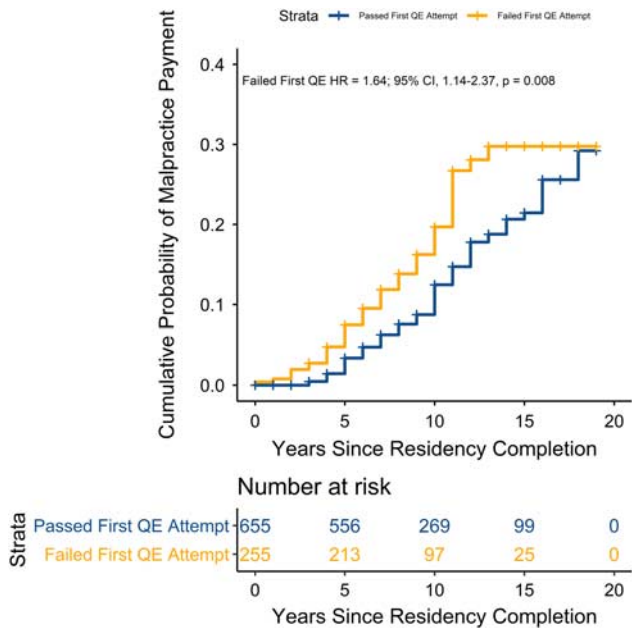


FIGURE 4. Cumulative incidence plot of medical malpractice payments over time by QE examination performance.

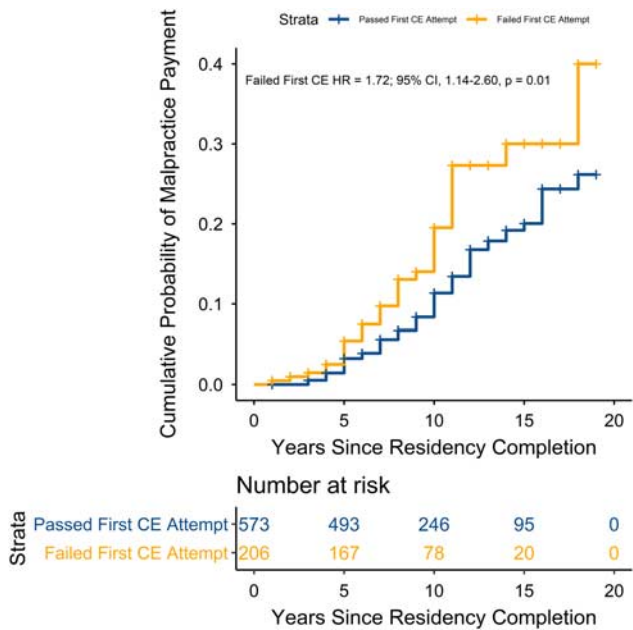


FIGURE 5. Cumulative incidence plot of medical malpractice payments over time by CE examination performance.

study add to the evidence that the multistep process for obtaining board certification has value.⁷

It is plausible to posit that the practice environments for board-certified surgeons are different than noncertified surgeons. Board certification is typically a component of the hospital credentialing process²⁹ and, as such, board-certified surgeons may have the opportunity to perform riskier procedures than noncertified surgeons. This may account for why the raw malpractice payment rates did not differ for board-certified and noncertified surgeons but were significantly lower for board-certified surgeons in the matched sample. It is also possible that board-certified surgeons would have better insurance than noncertified surgeons, which would reduce the likelihood that they would have a malpractice report against them. However, there is a lack of data that compares the practice characteristics of board-certified surgeons to noncertified surgeons.

One of the important strengths of this study is that it relied upon a propensity score-matched sample for the analysis with known variables. The methodology we used in this study accounted for any potential differences in practice volume or average patient risk. This should limit the potential for any bias in the results based on confounding factors between certified and noncertified surgeons. In addition, a sensitivity analysis that assumed a high error rate in matching between the ABS and NPDB data still found significantly higher malpractice payment rates in noncertified surgeons. Finally, retaining only severe injuries as an outcome variable did not change the findings of the study.

It is important to note that this study does not establish a causal link between board certification and malpractice payments. However, stakeholders who look for quality indicators in surgeons are unlikely to be concerned about causal relationships between those indicators and practice outcomes. This study does provide further evidence that board certification is an indicator of malpractice risk. While having a malpractice report does not directly indicate the quality of care, there is research that has

shown some evidence that there may be a link between medical malpractice payments with practice and patient outcomes.^{20–23}

Limitations

There are several limitations to this study. First, the reporting of a malpractice injury for an adverse event is not universal. These events depend upon the harmed patient filing a malpractice claim. A patient's likelihood of filing such a claim may be dependent upon individual factors, such as their socioeconomic status or communication issues with the provider.^{26,30} Other factors could include the state where the injury happened, as some states have very low medical malpractice caps.³¹ Moreover, insurance companies may opt to make medical malpractice payments to patients, even when no medical malpractice occurred. This can happen if an insurance company decides that settling out of court can be less costly or more efficient than navigating a potentially costly court case. However, we would not expect these calculations to systematically differ by whether a surgeon is a board-certified. While we found no differences in aggregate patient income levels for certified and noncertified surgeons, we could not control these factors at the individual patient level. In addition, our propensity score-matched sample was limited by known variables in the data to which we had access. Other factors that relate to the quality of care which were not included are teaching hospital status.³² It is possible that other control variables that were not examined could influence the findings of this study.

A further limitation may be the nature of the noncertified sample after matching on practice data. The malpractice payment rate in the full sample of noncertified surgeons was much lower than in the matched sample, indicating there are likely some systematic differences in noncertified surgeons who are available in the CMS data and those who are not. Notably, when we analyzed the full sample, there were no differences in malpractice payment claim rates between the 2 groups of surgeons. We strongly believe that the confounding variables need to be included when comparing these groups while noting the potential limitations of the observed differences.

Finally, we did not control for years in practice in the matched sample, meaning that the noncertified group may be younger and therefore have more claims. However, we did test the assumption of linearity of the relationship between board certification status and the rate of malpractice claims over time, and it was not violated. Therefore, we do not believe that time in practice would likely explain the observed differences in this study.

CONCLUSIONS

Future studies should investigate the relationship between board certification and practice and patient outcomes. Multiple studies have established a link between obtaining board certification and disciplinary actions, operative outcomes, and malpractice payments. Other studies could examine whether board certification links to outcomes in a larger number of procedures and also whether certification relates to patient safety indicators. Researchers should also continue to examine the relationship between maintaining board certification over the course of a surgeon's career and practice outcomes.

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